Pre-analysis Plan:

Impact of Having Daughters on Attitudes, Preferences, and Behaviors

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1 Motivation

Our analysis will provide evidence on the effect of having daughters on three main outcomes: (i) individuals' support for gender equality, (ii) political preferences, and (iii) equality in the household measured by the division of housework and childcare. In addition, by examining heterogeneity by parent gender, we will assess whether the impact of having daughters differs between mothers and fathers, particularly in relation to their support for gender equality.

2 Hypotheses

H1. *Having daughters leads to stronger support for gender equality (as measured by attitudes).*
That is, denoting X as a measure of support for gender equality,
(X for respondents with daughters) > (X for respondents without daughters).

H2. *Having daughters leads to more left-leaning political preferences.*
That is, denoting Y as a measure of left-leaning political preferences,
(Y for respondents with daughters) > (Y for respondents without daughters).

H3. *Having daughters leads to more gender equality in the household (as measured by behaviors or outcomes).*
That is, denoting Z as a measure of gender equality in the household,
(Z for respondents with daughters) > (Z for respondents without daughters).

H4. *Having daughters leads to a larger increase in the support of gender equality for men than for women (with gender equality measured by attitudes).*
That is, denoting X as a measure of support for gender equality,

$$\begin{pmatrix} X \text{ for male respondents with daughters} \\ -X \text{ for male respondents without daughters} \end{pmatrix} > \begin{pmatrix} X \text{ for female respondents with daughters} \\ -X \text{ for female respondents without daughters} \end{pmatrix}.$$

3 Data and variables

To analyze our hypotheses, we use data from the SOEP v39 (2022), a representative longitudinal study on households and individuals living in Germany. The survey has been conducted annually since 1984 (Goebel et al., 2019). The relevant population are persons living in private households in Germany. All samples of SOEP are multi-stage random

samples which are regionally clustered. The respondents (households) are selected by random-walk.

3.1 Dependent variables

3.1.1 Support for gender equality

We measure support for gender equality using a question from the SOEP that asks respondents to what extent they agree with the following statement: "It's best if the man and the woman work the same amount so they can share the responsibility for taking care of the family and household equally" (*plh0308_v2*). Responses are recorded on a 7-point Likert scale, ranging from one (disagree completely) to seven (agree completely). This question is available only in 2018. We construct a binary variable indicating agreement with the statement, coded as one if the response falls between five and seven, and zero otherwise.

3.1.2 Political preferences

To proxy parents' left-leaning political preferences, we rely on respondents' self-reported political attitudes. The SOEP asks individuals to place their political views on a scale from zero (far left) to ten (far right) in years 2005, 2009, 2014, and 2019 (*plh0004*). Based on this, we construct a binary indicator for left-leaning political preference that equals one if the respondent selects a value between zero and three, and zero otherwise.

3.1.3 Gender equality in the household

The SOEP provides information on the number of hours spent on certain activities on a typical weekday. We focus on housework and child care (*pli0043_h* and *pli0044_h*, available in all years). We apply the following formula to calculate an equality index (EI) in the household.

$$EI = 1 - \frac{|H_m - H_f|}{H_m + H_f}, \tag{1}$$

where H_m stands for the combined number of hours the mother spends on housework and child care while H_f refers to the number of hours the father spends on the same activities.

3.2 Main independent variables

Our main independent variable is a dummy variable indicating whether the individual has daughters. It takes the value of one if the individual has at least one daughter, and

zero if they have only sons. Information on the gender of the child is obtained from the *BIOBIRTH* dataset.

3.3 Control variables

We include the following control variables to the regression analysis.

- Age: We use the information on birth year (*gebjahr*) and survey year (*syear*) to generate the age of respondents.
- Gender: We generate a dummy variable for being a female using *sex*.
- Migration background: Using *migback*, we construct dummy variables for having no migration background, having a direct migration background, and having an indirect migration background.
- Education: We proxy education using the amount of education or training in years (*pgbilzeit*).
- Employment status: Using *pglfs*, we generate a dummy variable for being working.
- Age of the first child: We apply the birth year of the first child *kidgeb01* and the survey year (*syear*) to generate the age of the child.
- Survey year: Using *syear*, we construct dummy variables for the survey years.
- Living region (federal state): We employ *bula_h* to generate dummy variables for the 16 federal states in Germany.

4 Empirical strategy

4.1 Sample restrictions

To identify the causal impact of having daughters on gender equality attitudes, political preferences, and gender equality in the household, we restrict our analysis to households with only one child and assume that the child's gender is exogenously determined. We disregard the possibility of gender selection through assisted reproduction or abortion, as such practices are rare in Germany. In our baseline analysis, we apply the following sample restrictions:

1. Both parents are the biological parents of the child and reside in the same household.
2. Neither parent has children from previous relationships.

3. The couple has only one child together.

4.2 Empirical models

To measure the impact of having daughters on individuals' support for gender equality (**H1**), we estimate Ordinary Least Squares (OLS) models using the following regression specification:

$$X_{i,2018} = \alpha_0 + \alpha_1 Daughter_{i,2018} + \boldsymbol{\phi Control}_{i,2018} + \varepsilon_{i,2018}, \tag{2}$$

where $X_{i,2018}$ denotes the gender equality measure for individual i in 2018. The variable $Daughter_{i,2018}$ is a binary indicator for whether the individual i has a daughter in 2018. $\boldsymbol{Control}_{i,2018}$ is a vector of covariates that have been introduced in Section 3, including individual i's age, gender, migration background, years of education, employment status, child age, and living region. $\varepsilon_{i,2018}$ is the error term. Robust standard errors are applied. A positive coefficient of α_1 would provide evidence in support of the hypothesis.

To study the effect of having daughters on individuals' left-leaning political preferences (**H2**), we estimate OLS models using the following regression specification:

$$Y_{it} = \beta_0 + \beta_1 Daughter_{it} + \boldsymbol{\phi Control}_{it} + \lambda_t + \varepsilon_{it}, \tag{3}$$

where Y_{it} denotes the left-leaning political preferences of individual i in year t. The variable $Daughter_{it}$ is a dummy indicator for whether the individual i has a daughter in year t. $\boldsymbol{Control}_{it}$ is a vector of covariates for individual i in year t, defined as previously, including demographic and socioeconomic characteristics. λ_t captures the year fixed effects and ε_{it} is the error term. Standard errors are clustered at the individual level to account for within-person correlation over time. A positive coefficient of β_1 would provide evidence in support of the hypothesis.

To identify the impact of having daughters on individuals' gender equality in the household (**H3**), we estimate OLS models using the following regression specification:

$$Z_{it} = \gamma_0 + \gamma_1 Daughter_{it} + \boldsymbol{\phi Control}_{it} + \lambda_t + \varepsilon_{it}, \tag{4}$$

where Z_{it} denotes the measure of gender equality in the household of individual i in year t. The variables $Daughter_{it}$, $\boldsymbol{X}_{it}$, λ_t, and ε_{it} are defined the same as in equation (3). Standard errors are clustered at the household level to account for intra-household correlation in outcomes related to gender equality in the division of housework and childcare. A positive coefficient of γ_1 would provide evidence in support of the hypothesis.

To test the last hypothesis (**H4**), whether having daughters leads to a larger increase in the support of gender equality for men than for women, we estimate OLS models using the following regression specification:

$$X_{i,2018} = \alpha_0 + \alpha_1 Daughter_{i,2018} + \alpha_2 Male_i + \alpha_3 Daughter_{i,2018} \times Male_i + \boldsymbol{\phi Control}_{i,2018} + \varepsilon_{i,2018}. \quad (5)$$

Here, $X_{i,2018}$ denotes the outcome variable measuring support for gender equality for individual i in 2018, using the same measures as employed in the analysis of **H1**. $Male_i$ is a dummy variable indicating whether individual i is male. All control variables are the same as in equation (2) expect for gender. Robust standard errors are applied. A positive coefficient of α_3 would suggest that the impact of having daughters on support for gender equality is stronger for men than for women.

5 Conclusion

This pre-analysis plan outlines our empirical strategy to investigate the causal impact of having daughters on individuals' support for gender equality, political preferences, and gender equality in the household. We aim to provide credible evidence on how child gender shapes parental attitudes and behavior. We specify hypotheses, define variables, and outline identification strategies to ensure transparency and rigor. As output of our analysis, we will present four separate tables corresponding to each of the four hypotheses. This plan serves as a commitment to a predefined analytical framework and contributes to the credibility and replicability of our research.

References

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